

Om Jadhav

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PROFESSIONAL SUMMARY

GenAI Engineer with hands-on experience designing, building, and deploying LLM-powered applications, Retrieval-Augmented Generation systems, and embedding-based semantic search solutions. Strong background in Python, FastAPI, vector databases, and orchestration frameworks such as **LangChain and Agno**. Experienced in developing scalable, production-ready GenAI services with prompt engineering, evaluation workflows, structured logging, and backend integrations across enterprise-style systems.

EDUCATION

Savitribai Phule Pune University	Pune, Maharashtra
<i>Bachelor of Engineering in Information Technology</i>	<i>Graduated: June 2026</i>

EXPERIENCE

Parenthese Pvt.Ltd | GenAI / AI Engineer Pune ,Maharashtra | June 2025 – Nov 2025

- Built and deployed GenAI applications including LLM-powered assistants and automation workflows
- Engineered semantic search pipelines with pgvector (PostgreSQL) and all-MiniLM-L6-v2 embeddings, implementing hybrid retrieval (BM25 + vector) achieving 92% relevance accuracy
- Developed FastAPI-based backend services to expose GenAI capabilities as scalable REST APIs
- Implemented prompt iteration, testing, and fallback logic to improve response reliability and accuracy
- Added structured logging, error handling, and validation to support production-grade AI workflows

PROJECTS

Enterprise Agentic CNC Automation System | [Link](#)

Tools: FastAPI, MCP Server,Agno, PostgreSQL, Gemini LLM | Sep 2025

- Designed an AI-driven orchestration layer using LangChain to automate CNC operational tasks and decision workflows.
- Implemented **role-based agent collaboration** for data retrieval, reasoning, and command execution.
- Integrated **Gemini LLM** for contextual decision-making and explainable outputs, ensuring safe agent operations.
- Deployed modular backend via **FastAPI and PostgreSQL**, enabling scalable, autonomous workflows.

Disease Outbreak & Vaccination WhatsApp Chatbot System | [Link](#)

Tools: FastAPI, Twilio WhatsApp API, PostgreSQL, HTML,Agentic AI | Sep 2025

- Built an **AI-powered WhatsApp chatbot** for broadcasting disease alerts and vaccination updates based on user regions.
- Integrated **Twilio WhatsApp API** for two-way communication, enabling appointment bookings through chat.
- Implemented a **dynamic step-based workflow** with FastAPI backend for data collection and automation.

Mental Health Agent | [Link](#)

Tools: Next.js, FastAPI, Google Gemini LLM, RAG, Sentence-Transformers, PostgreSQL, gTTS, Speech Recognition | Nov 2025

- Developed empathetic AI therapist with RAG over DSM-5/CBT corpora, contextual session memory (PostgreSQL), and multi-turn conversation management.
- Implemented voice pipeline (Whisper STT → Gemini reasoning → gTTS) with <3s end-to-end latency; built semantic caching with FAISS reducing API costs by 30% .
- Designed secure authentication (JWT/OAuth2), encrypted journaling, and PII redaction pipelines.

ACHEIVEMENTS

- NVIDIA Certification** — *Building RAG Agents with LLMs* (2025)
- Anthropic MCP Advanced Topics Certification** — Model Context Protocol & LLM Orchestration (2025)
- Altire Data Science Master Internship** — Applied AI & Machine Learning projects (2025)

SKILLS

- Programming:**Python, JavaScript, TypeScript , Next.js, Node.js
- RAG & Vector Systems:** RAG pipeline architecture, pgvector, FAISS, Sentence-Transformers, Chunking strategies
- Agentic AI & LLMs:** LangChain, Agno, LangGraph, MCP (Model Context Protocol), ReAct, Function Calling
- Backend & Data:**REST APIs, Microservices, PostgreSQL, MySQL, SQL
- Engineering Practices:**Structured logging, error handling, API testing & debugging, Git version control
- Cloud & Deployment (Familiarity):**AWS , Docker